Solving multi-label problems with st-min-cut

Overview

Polynomial solvability

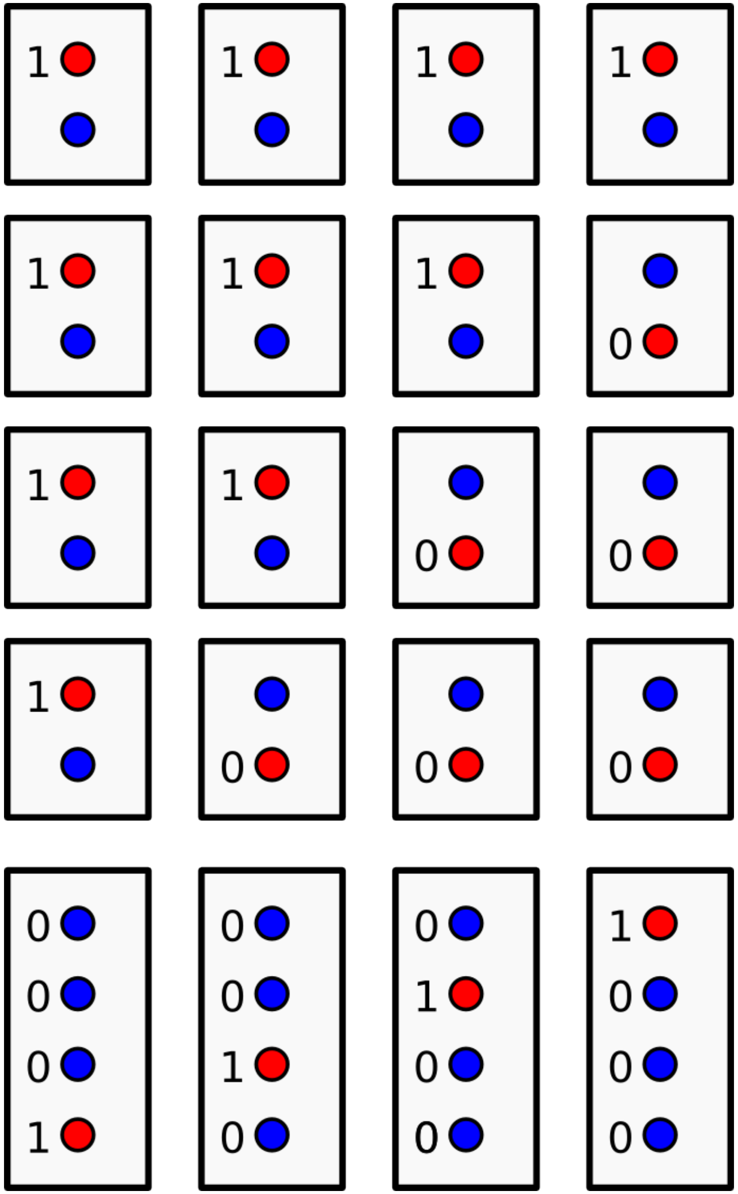
Min-cut

Submodularity

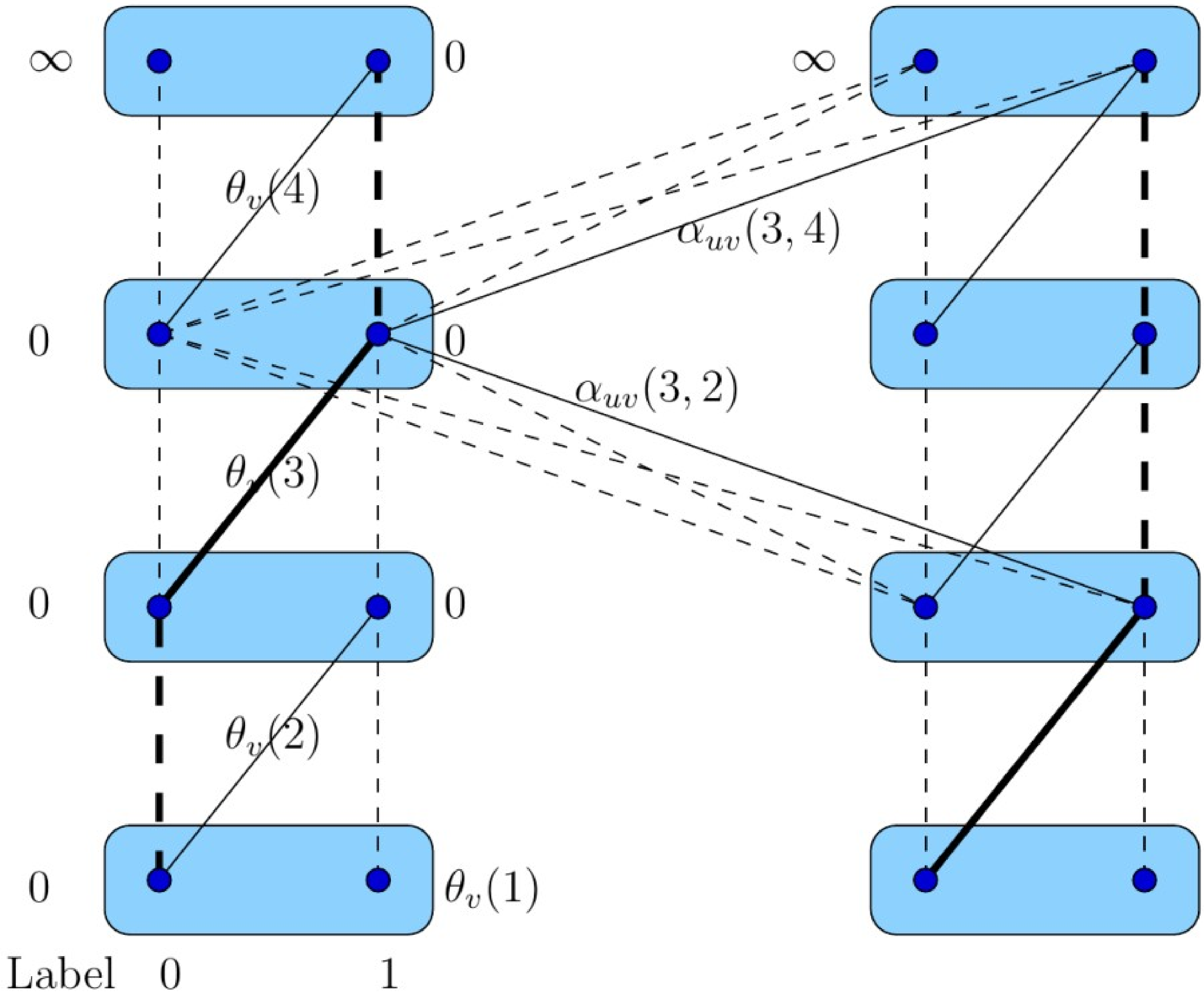
Non-negative edge costs

Binary energy

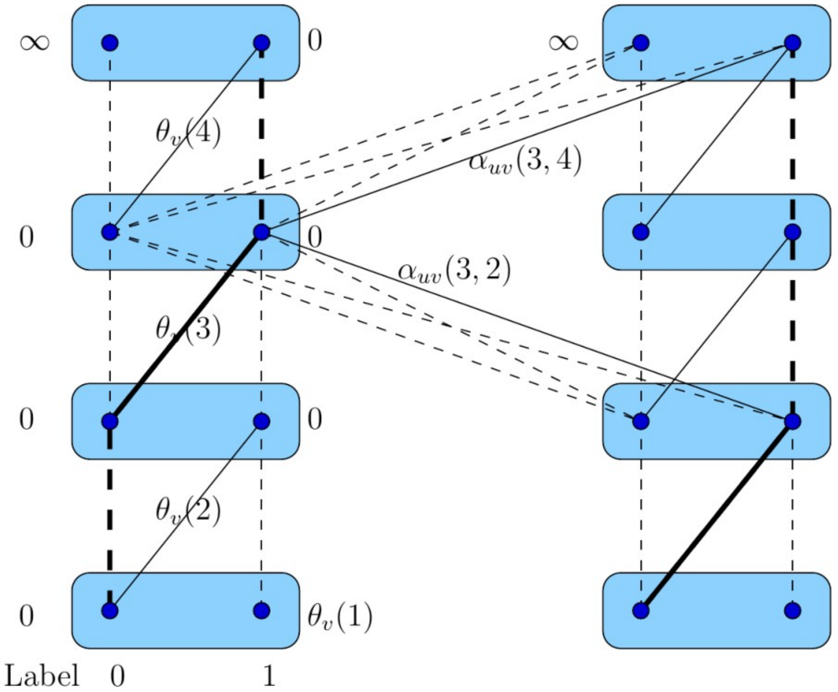
Transformation multilabel-to-binary graphical model

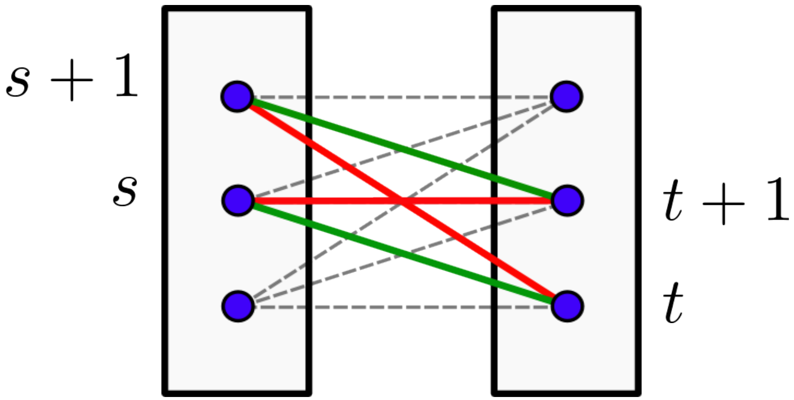


Transformation multilabel-to-binary graphical model



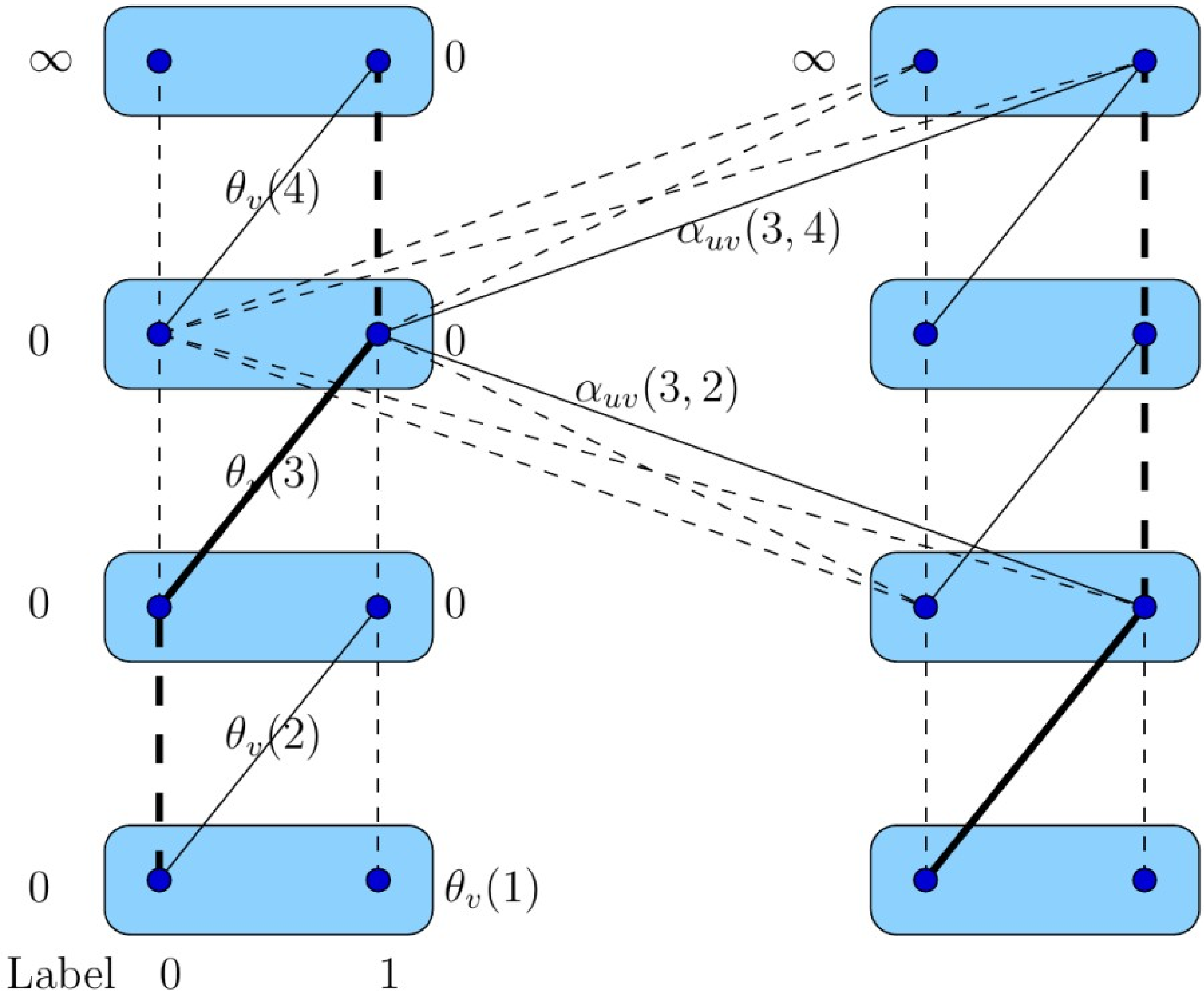
Transformation multilabel-to-binary graphical model



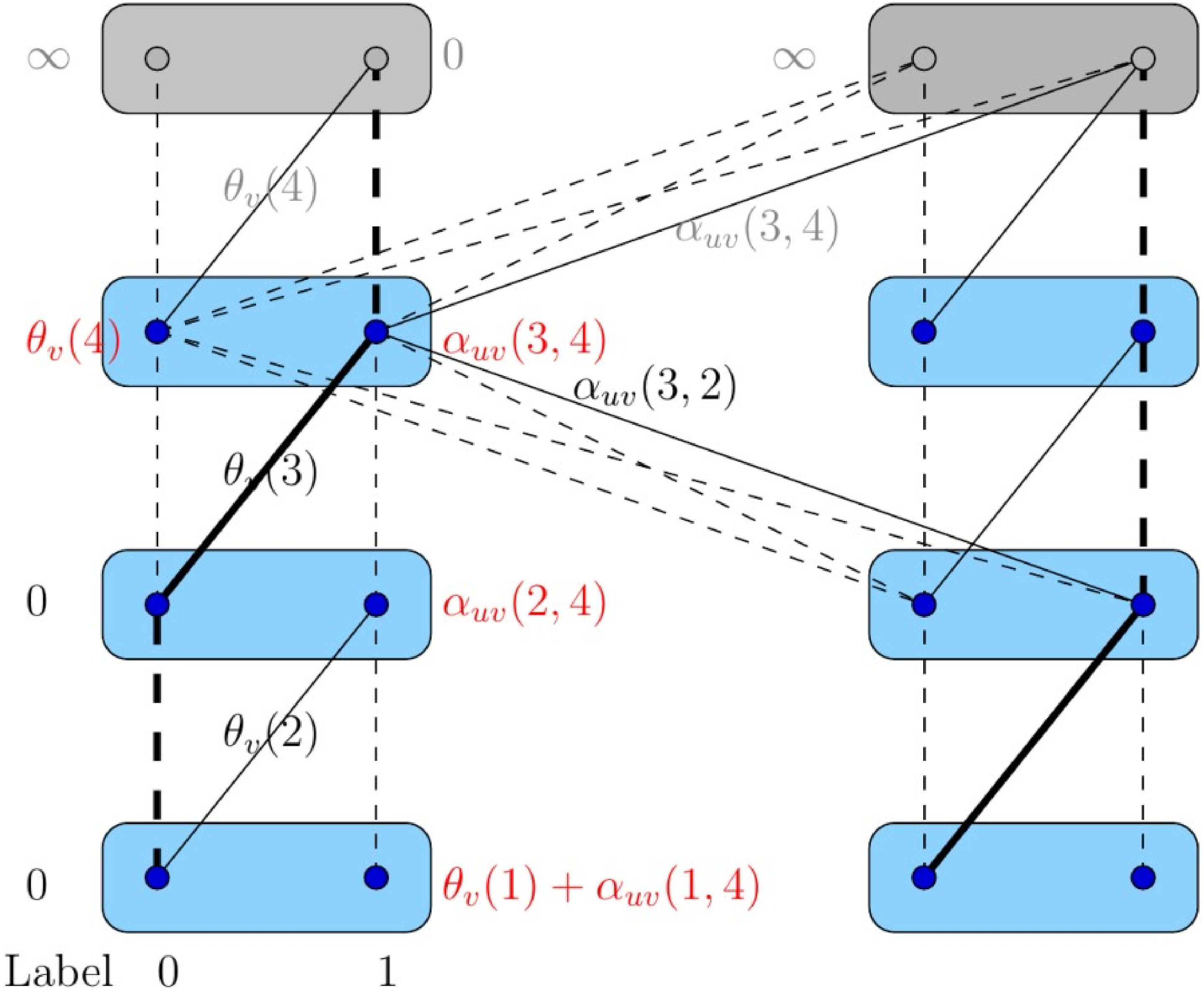


$$\alpha_{uv}(s,t) = \color{green}{\theta_{uv}(s,t) + \theta_{uv}(s+1,t+1)} - \color{red}{\theta_{uv}(s,t+1) - \theta_{uv}(s+1,t)}$$

Transformation multilabel-to-binary graphical model



Transformation multilabel-to-binary graphical model



Transformation multilabel-to-binary graphical model

Multi-label pairwise energy $\Leftrightarrow$ Binary pairwise energy

Transformation multilabel-to-binary graphical model

Multi-label pairwise energy	⇔	Binary pairwise energy
Submodular Multi-label pairwise energy	⇔	Submodular Binary pairwise energy

Transformation multilabel-to-binary graphical model

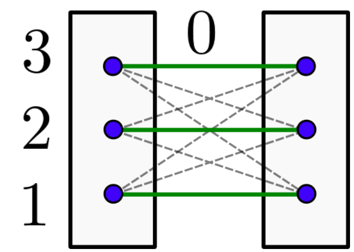
Multi-label pairwise energy	⟺	Binary pairwise energy
Submodular Multi-label pairwise energy	⟺	Submodular Binary pairwise energy
Min-Cut with non-negative edge weight	⟺	Submodular Binary pairwise energy

Transformation multilabel-to-binary graphical model

Multi-label pairwise energy	⇔	Binary pairwise energy
Submodular Multi-label pairwise energy	⇔	Submodular Binary pairwise energy
Min-Cut with non-negative edge weight	⇔	Submodular Binary pairwise energy
Min-Cut with non-negative edge weight	⇔	Submodular Multi-label pairwise energy

However, many important pairwise factors are non-submodular:

Truncated-convex potentials: $\mathcal{X}_v = \{1, \dots, n\}$



$a > 0$

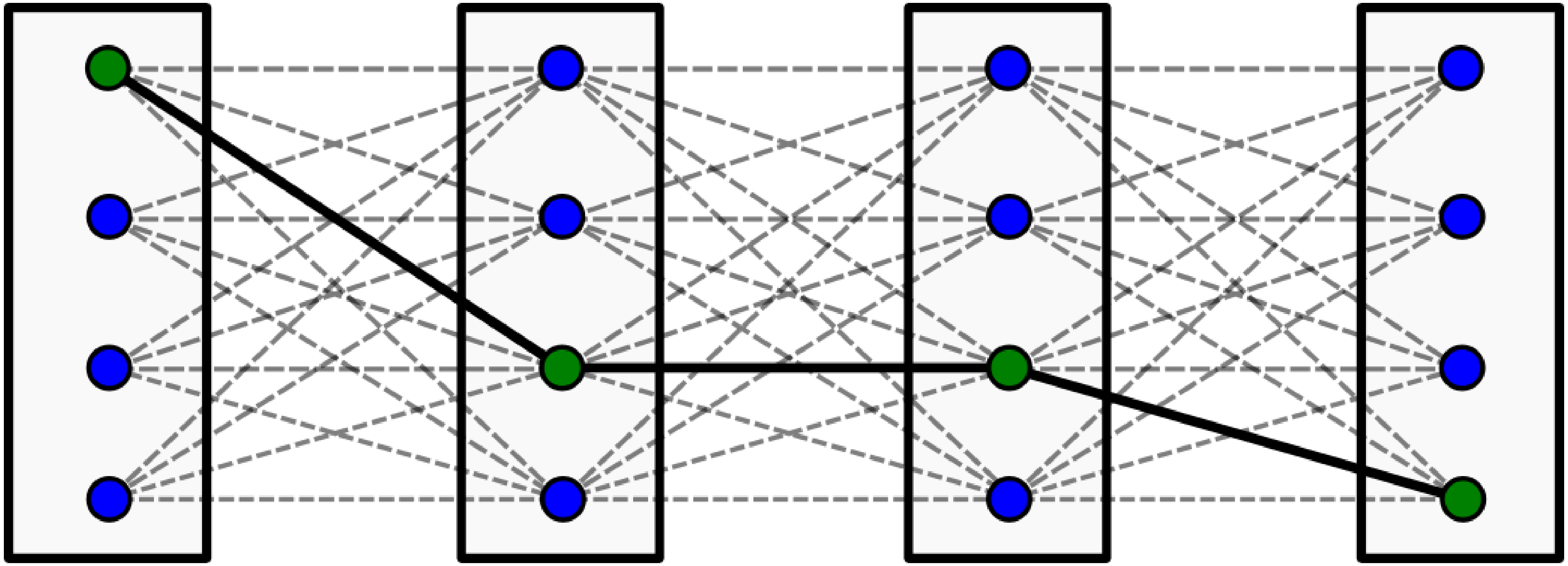
$$\theta_{uv}(s,t) = \min\{a, |s-t|\}$$

$$\theta_{uv}(s,t) = \min\{a, (s-t)^2\}$$

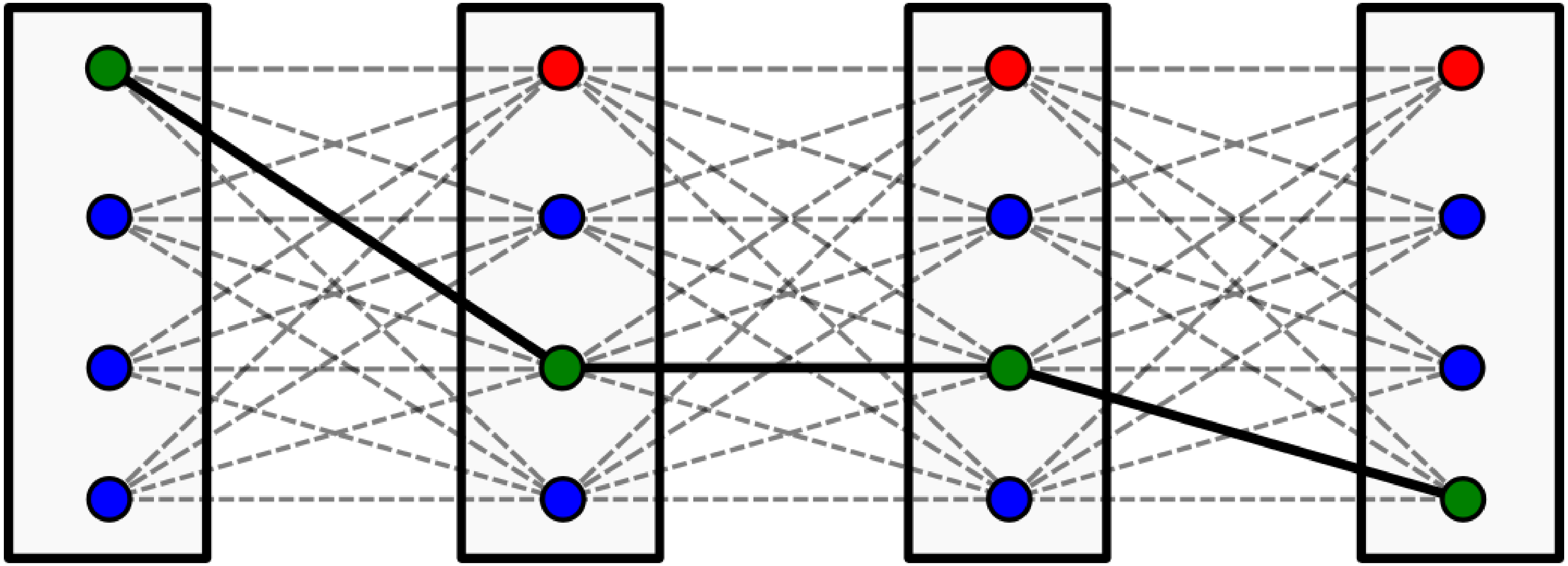
$$\theta_{uv}(s,t) = \min\{a, f(|s-t|)\}, \ \ f - \text{convex}$$

Alpha-Expansion Algorithm

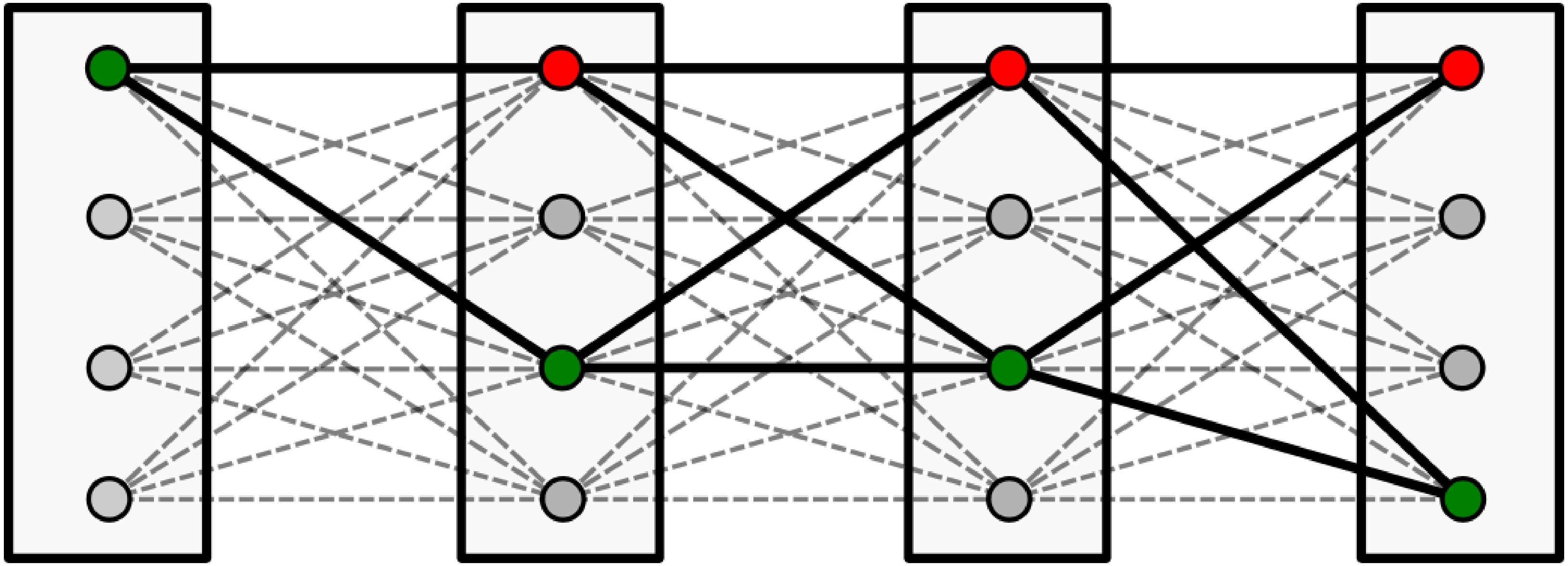
Select initial labeling x^0



Select α to expand

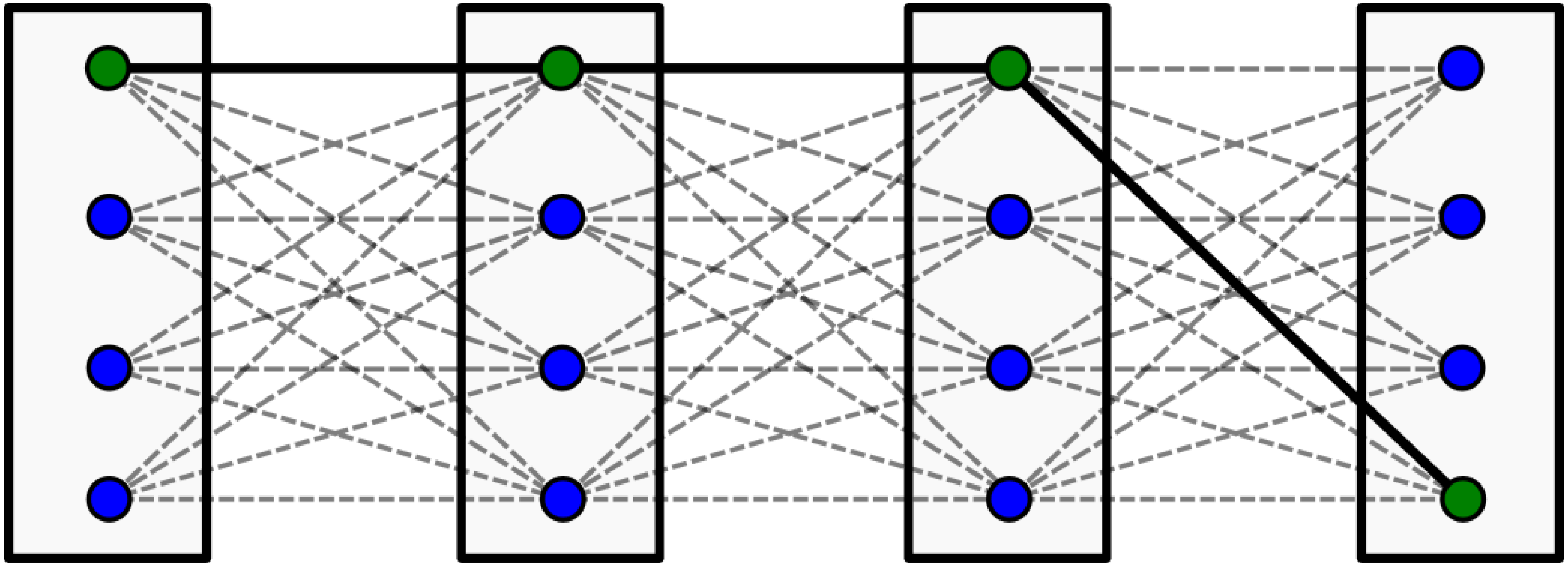


Solve the binary problem



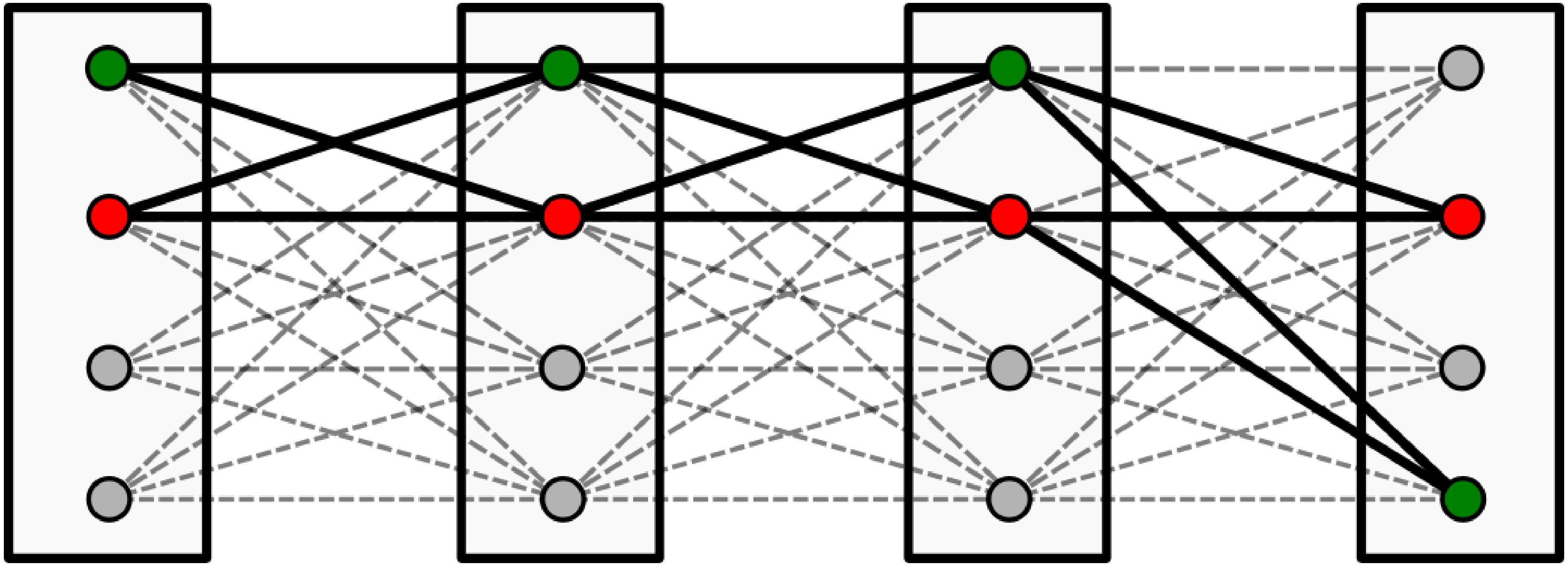
$\alpha = 1$

Turn solution to the current labeling



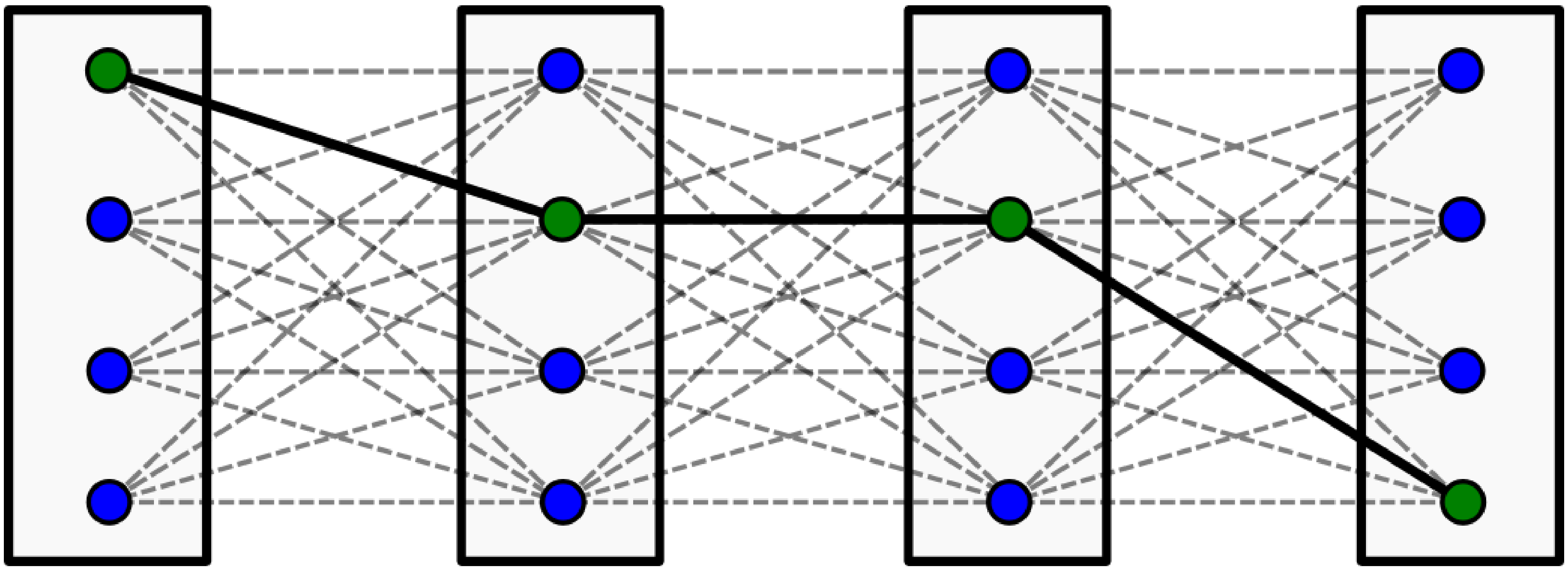
$$E(x^1) \leq E(x^0)$$

Select α to expand, solve the binary problem



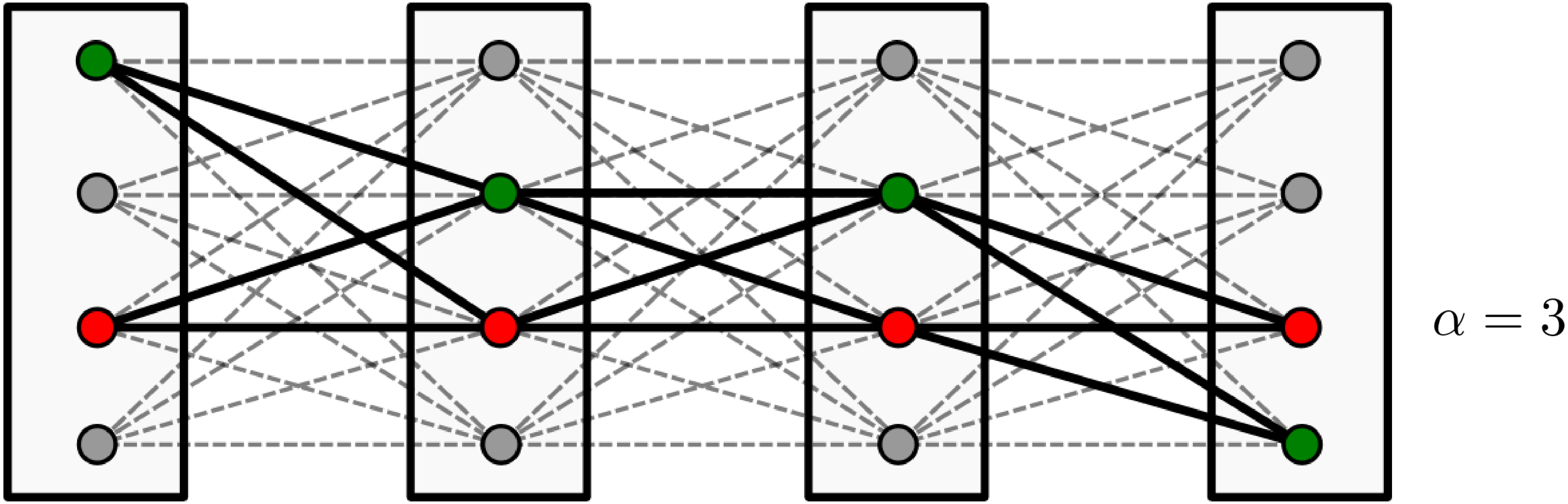
$\alpha = 2$

Turn solution to the current labeling

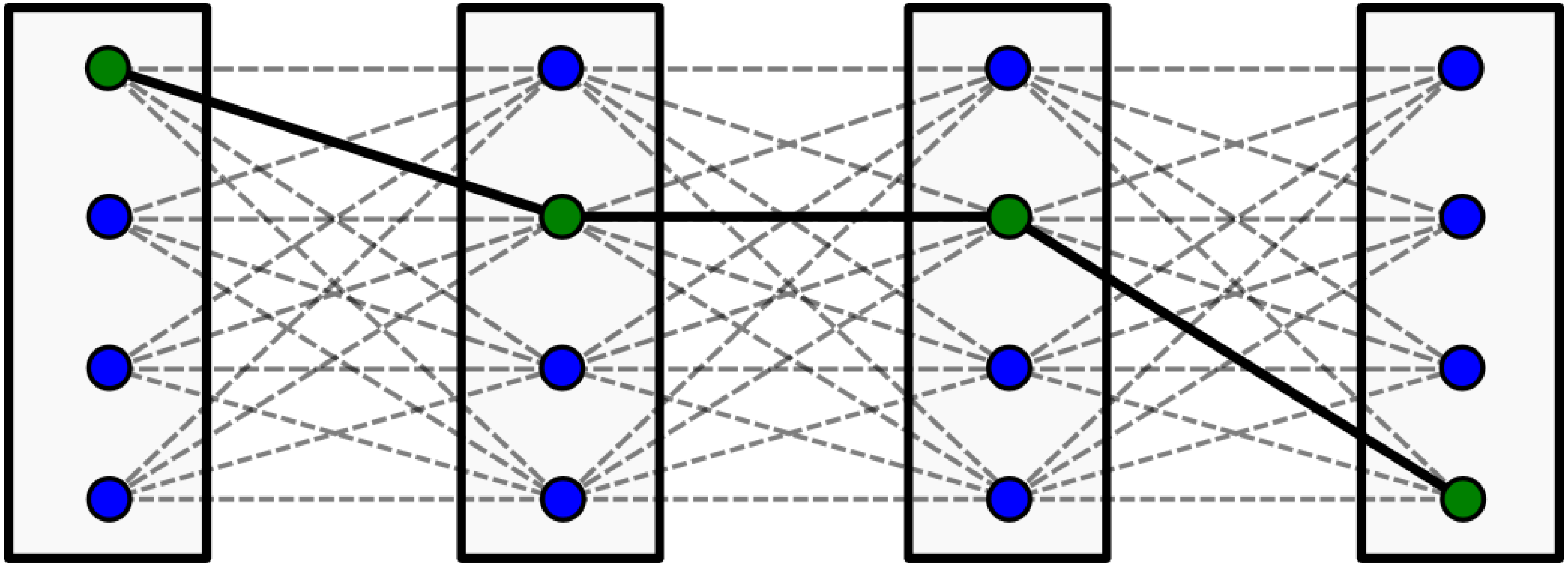


$$E(x^2) \leq E(x^1)$$

Select α to expand, solve the binary problem

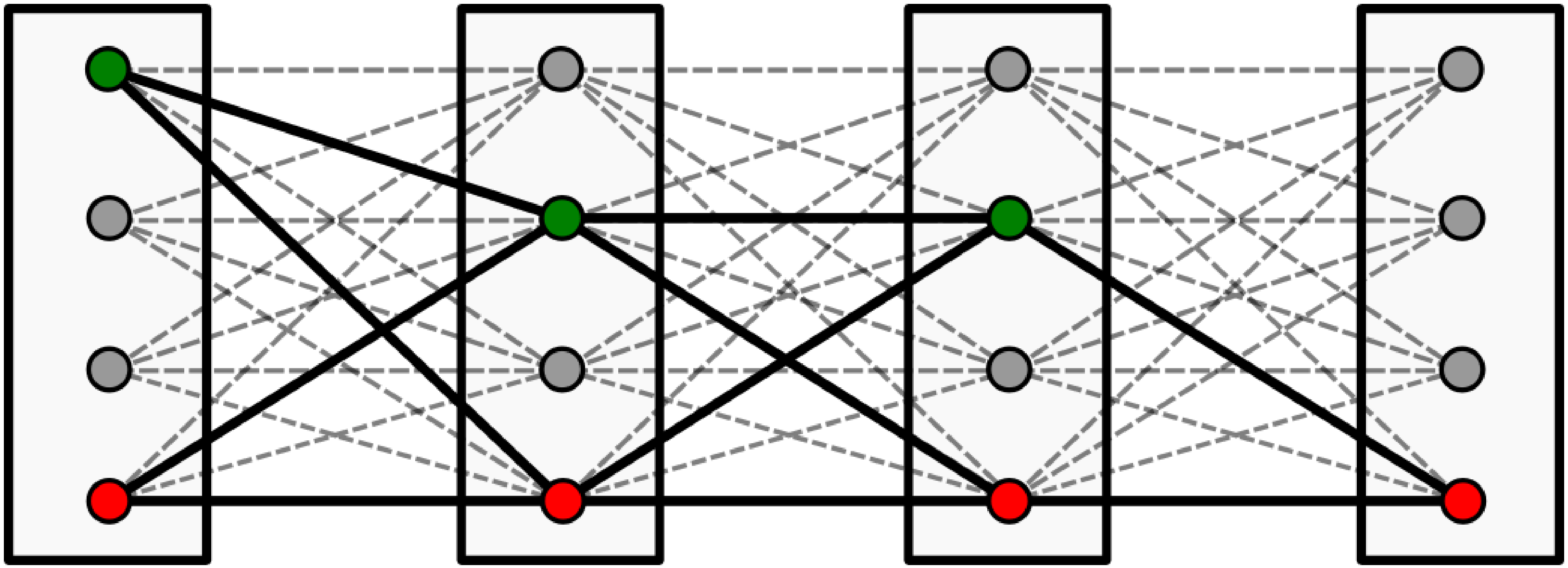


Turn solution to the current labeling



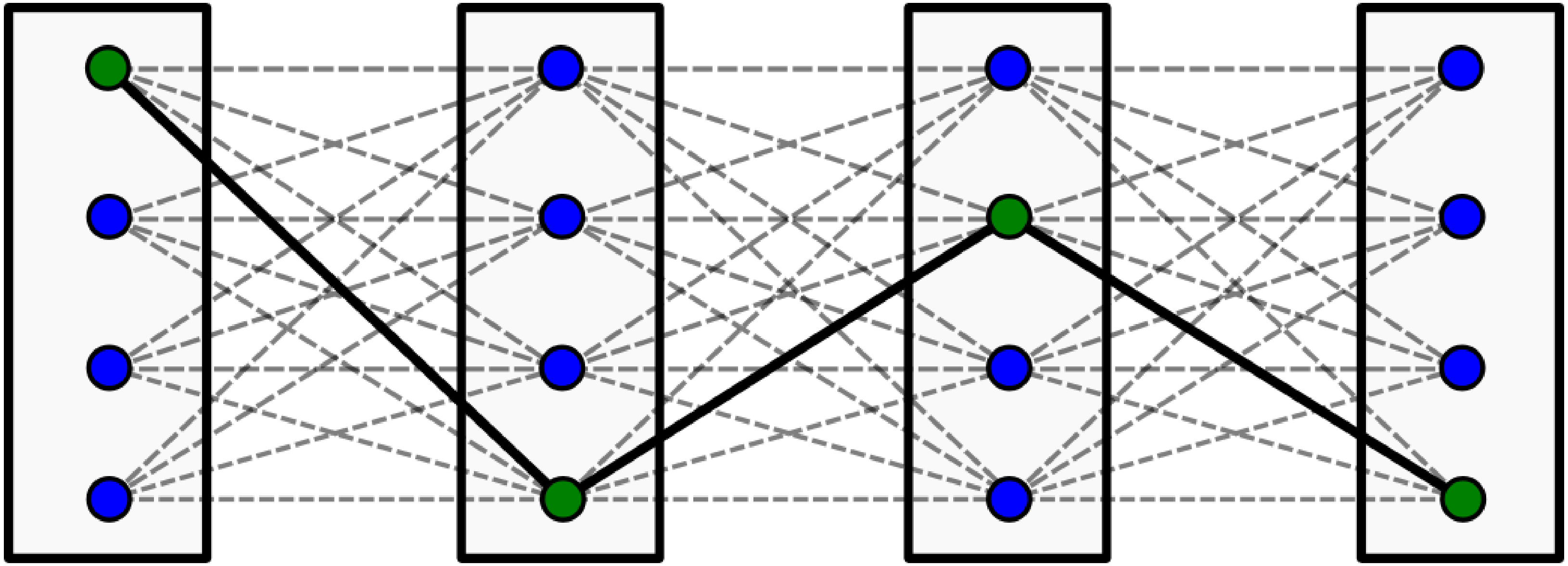
$$E(x^3) \leq E(x^2)$$

Select α to expand, solve the binary problem



$\alpha = 4$

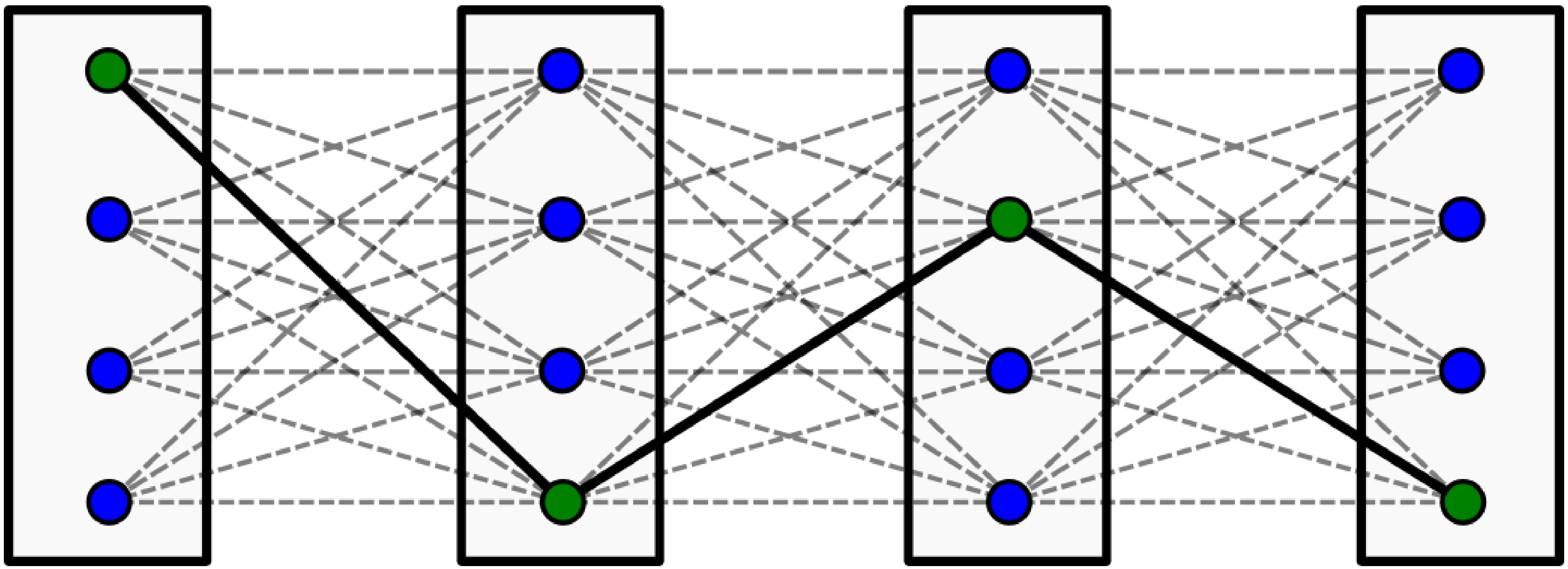
Turn solution to the current labeling



$$E(x^4) \leq E(x^3)$$

Alpha-Expansion Algorithm

Repeat until $E(x^t) = E(x^{t-L})$



Monotonicity: $E(x^{t+1}) \leq E(x^t)$

The binary subproblems are submodular, if pairwise potentials are:

- submodular

  or

- ?

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Definition: *f* - semi-metrics, if:

$$\text{(i)}\ f(x,x') \geq 0\ \text{(ii)}\ f(x,x') = 0 \text{ iff } x = x'\ \text{(iii)}\ f(x,x') = f(x',x).$$

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Definition: Let *f* be semi-metrics. It is metrics, if additionally:

$$f(x,x'') \leq f(x,x') + f(x',x'')$$

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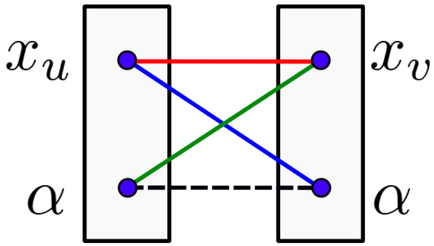

$$\theta(\alpha,\alpha) + \theta(x_u,x_v) \quad ? \quad \theta(x_u,\alpha) + \theta(\alpha,x_v)$$
$$0 \quad + \theta(x_u,x_v) \quad ? \quad \theta(x_u,\alpha) + \theta(\alpha,x_v)$$

Alpha-Expansion Algorithm: Analysis

Monotonicity: $E(x^{t+1}) \leq E(x^t)$

The binary subproblems are submodular, if pairwise potentials are:

- submodular

  or

- metric

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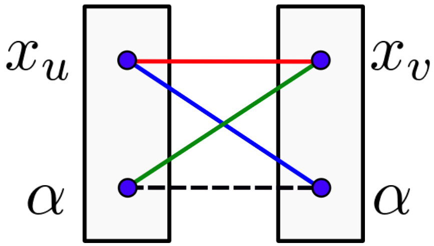
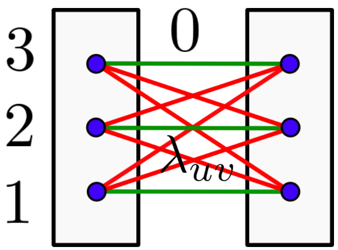


$$\theta(\alpha,\alpha) + \theta(x_u,x_v) \leq \theta(x_u,\alpha) + \theta(\alpha,x_v)$$
$$0 + \theta(x_u,x_v) \leq \theta(x_u,\alpha) + \theta(\alpha,x_v)$$

Potts model: $\theta_{uv}(s,t) = \lambda_{uv}[\![s \neq t]\!]$

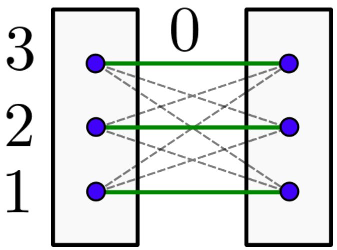


Are Potts potentials a metrics?

1) yes for $\lambda_{uv} > 0$
2) yes, for $\lambda_{uv} < 0$
3) no, because the triangle inequality is not satisfied
4) none is correct
5) no idea

Absolute difference: $\mathcal{X}_v = \{1, \ldots, n\}$

$$\theta_{uv}(s,t) = |s-t|$$



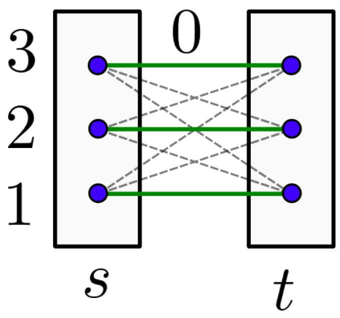
Are absolute-difference potentials metrics?

1) yes
2) no, because the condition f(x,x)=0 is not satisfied
3) no, because the triangle inequality is not satisfied
4) none is correct
5) no idea

Metrics: Examples

Truncated absolute difference: $\mathcal{X}_v = \{1, \ldots, n\}$

$$a > 0 \qquad \theta_{uv}(s,t) = \min\{a, |s-t|\}$$



Exercise: show that the truncated-absolute difference potentials are metrics

Metrics vs. Submodularity

Are metrics always submodular? (Hint: consider Multi-label Potts model)

1) yes, in particular the multi-label Potts model is metrics and submodular
2) no, the multi-label Potts is metric, but not submodular
3) yes, the multi-lable Potts is a bad example, because it is not a metrics
4) no, the multi-lable Potts is a bad example, because it is not a metrics
5) nether is correct
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Metrics vs. Submodularity

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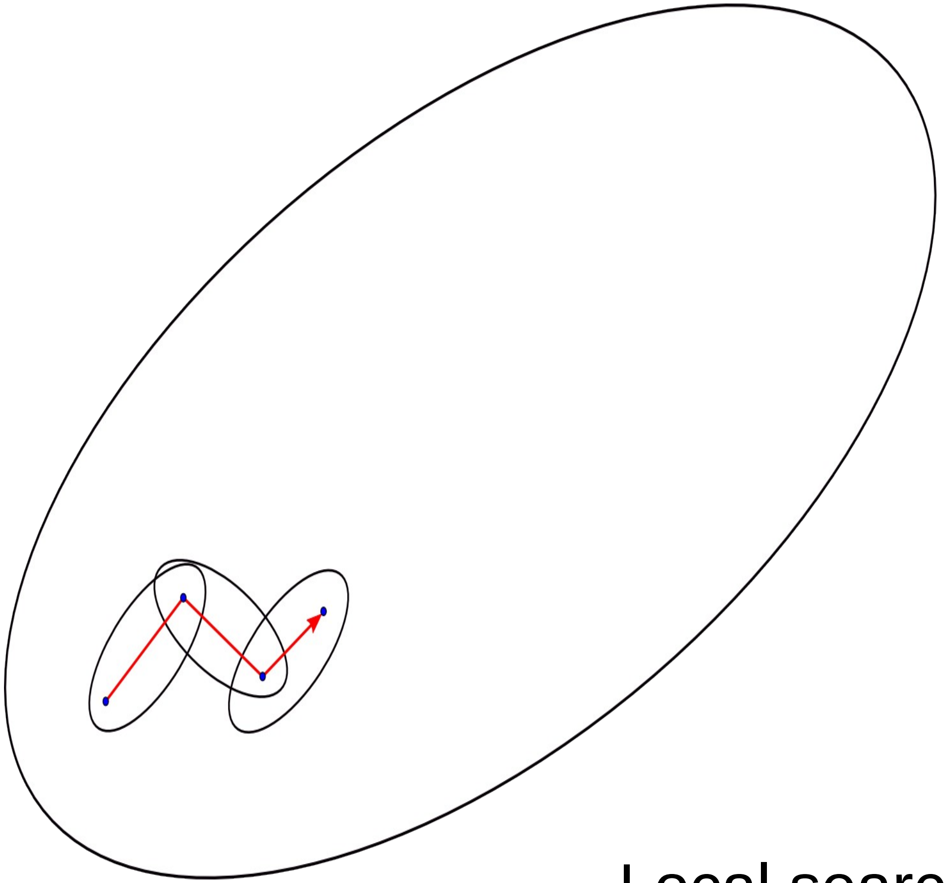
4) no, the multi-lable Potts is a bad example, because it is not a metrics
5) nether is correct
6) no idea

Are submodular pairwise potentials necessarily metrics?
(Hint: consider the condition f(x,x)=0)

1) yes, in particular the condition is fulfilled for submodular potentials
2) no, the condition need not be fulfilled by submodular potentials
3) no, adding a constant to all values does not change submodularity
4) nether is correct
5) answers 2) and 3) are both correct
6) no idea



Local search examples:

- coordinate descent

- gradient descent with line minimization

- simplex method for LPs

- alpha-expansion

Local search = Move-Making

Alpha-Expansion Algorithm: Optimality Bounds

Theorem: Let θ_{uv} be metrics and
*x** be an optimal labeling, *x′* be obtained by α-expansion. Then:

$$\sum_{v\in\mathcal{V}} \theta_v(x_v^*) + 2c \sum_{uv\in\mathcal{V}} \theta_{uv}(x_u^*, x_v^*) \geq \sum_{v\in\mathcal{V}} \theta_v(x_v') + \sum_{uv\in\mathcal{V}} \theta_{uv}(x_u', x_v')$$

where $$c := \max_{uv\in\mathcal{E}} \frac{\max_{x_v\neq x_u} \theta_{uv}(x_u, x_v)}{\min_{x_v\neq x_u} \theta_{uv}(x_u, x_v)}$$

Potts: c=1
Truncated linear and quadratic: c=a (for a>1)
Linear and quadratic: c=L-1 (L =number of labels)

In practice, the algorithm typically performs much better

Even the metrics conditions are often too strict

Unary factors =0

$$\theta_{uv}(x_u, x_v) = \begin{cases} 0, & x_u = x_v \\ |\mathrm{Color}_u(x_u) - \mathrm{Color}_v(x_v)|, & x_u \neq x_v \,. \end{cases}$$

Image credit: Richard Szeliski et al.. A comparative study of energy minimization methods for markov random fields with smoothness-based priors. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2008.

Even the metrics conditions are often too strict

Unary factors =0

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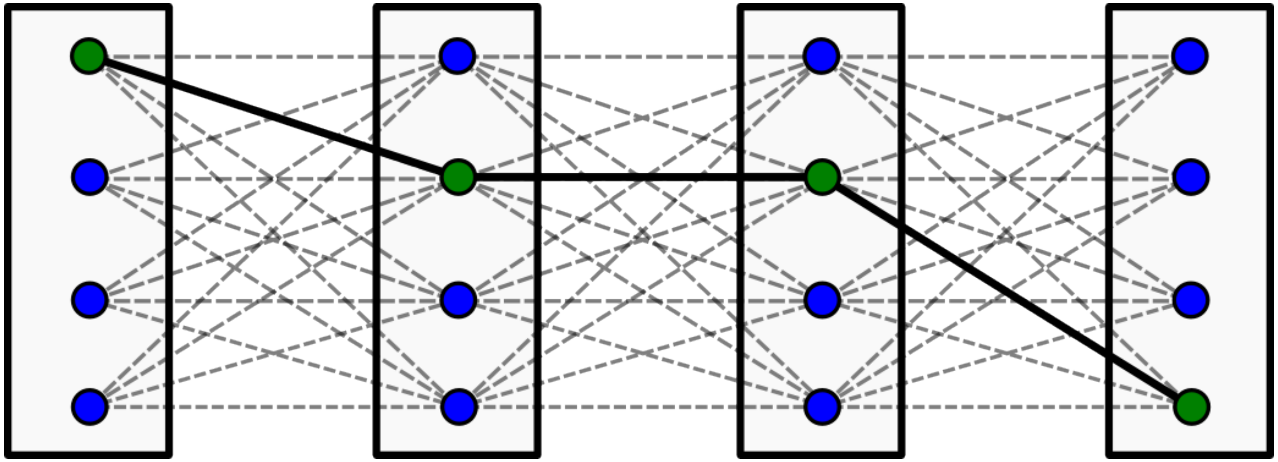
The pairwise factors form:

1) a metrics
2) a semi-metrics
3) are submodular
4) none of the above
5) no idea

Image credit: Richard Szeliski et al.. A comparative study of energy minimization methods for markov random fields with smoothness-based priors. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2008.

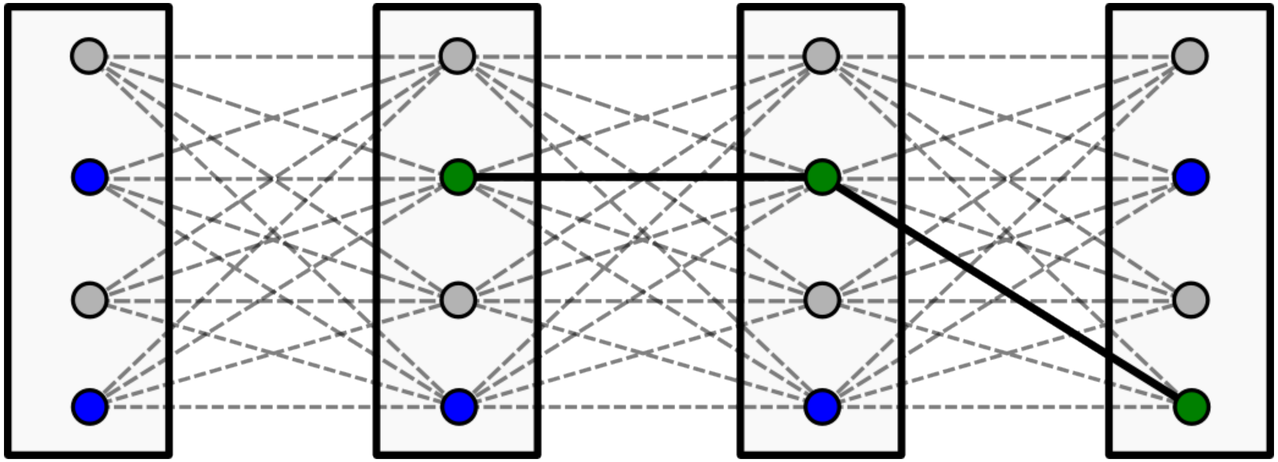
Alpha-Beta-Swap Algorithm

Select initial labeling x^0



Alpha-Beta-Swap Algorithm

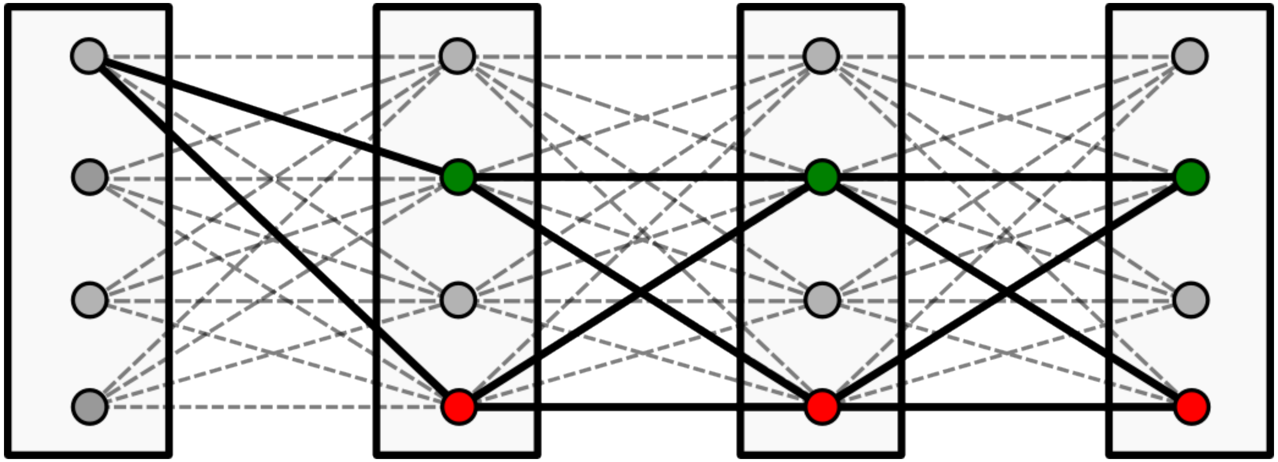
Select α and β to swap



$\alpha = 2$

$\beta = 4$

Solve the binary problem



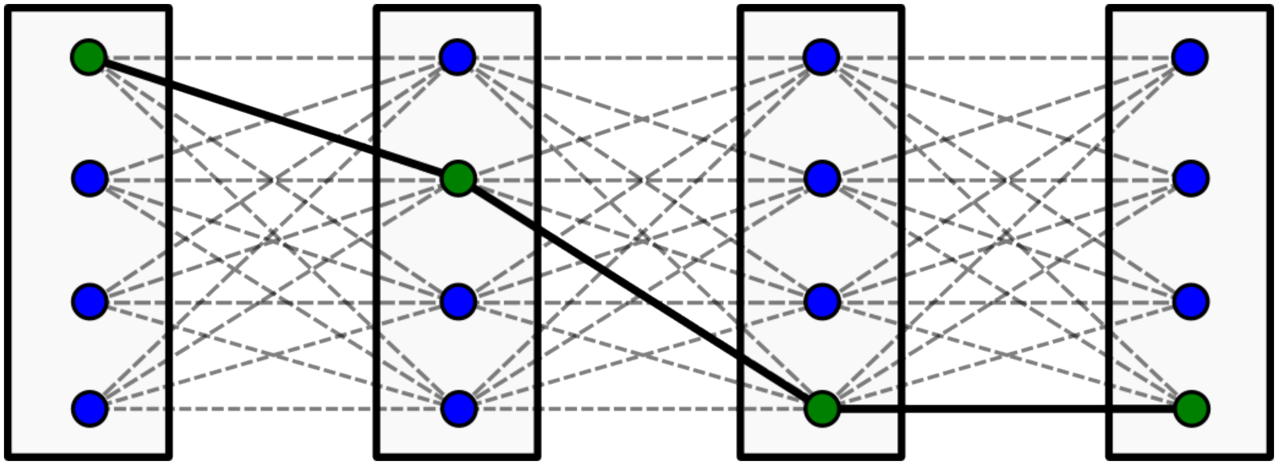
$\alpha = 2$

$\beta = 4$

Turn solution to the current labeling

Repeat until $E(x^t) = E(x^{t-\binom{L}{2}})$

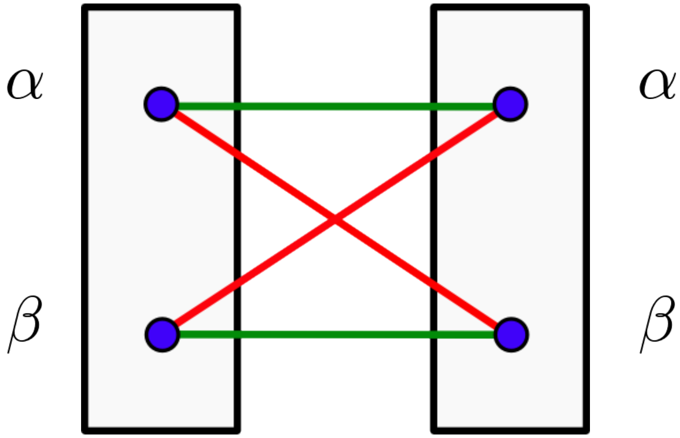
On each iteration of the algorithm does $\binom{L}{2} = O(L^2)$ swaps



$\alpha = 2$

$\beta = 4$

Alpha-Beta-Swap Algorithm: Analysis



If θ is a semi-metrics:

$$\theta(\alpha,\alpha) < \theta(\alpha,\beta)$$

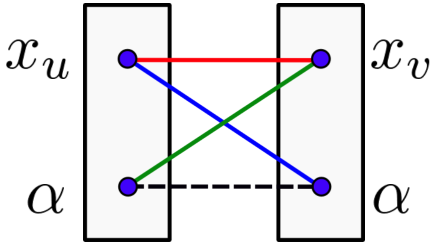
$$\theta(\alpha,\alpha) + \theta(\beta,\beta) < \theta(\alpha,\beta) + \theta(\beta,\alpha)$$

No optimality bounds. The algorithm can be arbitrary bad.

See example in the script.

MRF Middlebury Benchmark: http://vision.middlebury.edu/MRF/

Alpha-expansion: truncate $\theta(x_u, x_v)$ to force the inequality:



$$0 \; + \theta(x_u, x_v) \;\; \leq \;\; \theta(x_u, \alpha) + \theta(\alpha, x_v)$$

Still preserves monotonicity of the algorithm.

Use cases for pairwise potentials:

Multilabel submodular – either min-cut or alpha-expansion
Metric – alpha-expansion
Semi-metric – alpha-beta-swap or “truncated” alpha-expansion

Outlook:

How to deal with general pairwise potentials?